Old Exam Questions
AI in Life Sciences

Worked Solutions

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How this file is organized

This is the **solutions companion** to `exams.tex`. Every question is followed *immediately* by a green **answer box**. Inside it, each option is marked ✓ (correct) or × (incorrect), with a short justification — *why* the correct options are correct and, for the tricky / near-miss distractors, *why* they are wrong. Questions come from two sittings: the **2024** exam (25 June 2024, Sections 1–6, *multiple-select* — one *or more* options correct) and the **2026** exam (25 June 2026, Sections 7–10, *single-best-answer* — exactly one option correct). Wording and the answer options have both been paraphrased (the meaning and the correct answers are preserved). For a clean practice set with blank answers, use `exams.tex`.

Contents

Drug Discovery, Molecules & Representations

Question 1: SMILES strings

Regarding the SMILES notation for molecules, which of the following statements hold?

- ☐ SMILES is short for the *Simplified Molecular Input Line Entry System.*
- ☐ A SMILES encodes a molecule as a one-dimensional sequence of characters.
- ☐ A SMILES string cannot be turned into a molecular graph.
- ☐ Each molecule has exactly one SMILES, so no two distinct molecules can share a string.

Answers

- ✓ **A.** It is exactly the *Simplified Molecular Input Line Entry System.*
- ✓ **B.** A molecule is encoded as a 1D string of characters (atoms, bonds, ring/branch tokens).
- × **C.** SMILES *can* be parsed into a molecular graph — that round-trip is one of its main uses.
- × **D.** Not unique: a single molecule has *many* valid SMILES. Only a *canonical* SMILES (a fixed algorithm) gives one representative string — the plain format is not inherently unique.

Question 2: Reading a SMILES string

Consider the molecule *pyruvic acid* (2-oxopropanoic acid), i.e. a methyl group followed by a ketone carbonyl and then a carboxylic acid (CH_3–C(=O)–C(=O)–OH). Which of the SMILES strings below correctly encode this molecule?

- ☐ `O=C(C(=O)O)C`
- ☐ `CC(=O)C(=O)O`
- ☐ `CCCOOO`
- ☐ `OCCOO`

Answers

- ✓ **A.** `O=C(C(=O)O)C`: a carbonyl carbon bearing a `C(=O)O` (carboxyl) and a `C` (methyl) — exactly CH_3–CO–COOH, just written starting from the ketone.
- ✓ **B.** `CC(=O)C(=O)O`: methyl, ketone carbonyl, carboxyl read left-to-right — the same molecule.
- × **C.** `CCCOOO` is a chain of three carbons then three oxygens (no double bonds, no carboxyl) — not pyruvic acid, and chemically nonsensical.
- × **D.** `OCCOO` encodes a short O/C chain with the wrong connectivity and no ketone/carboxyl — not the target.

Question 3: Molecular fingerprints

Which of the following statements about molecular fingerprints are accurate?

- ☐ A unique molecular graph can be reconstructed from them.
- ☐ They are binary or numeric encodings of molecular structure.
- ☐ They serve similarity searches and machine-learning tasks.
- ☐ They give a 3D representation of the molecule.

Answers

- × **A.** They are *lossy* and often hashed (bit collisions), so you cannot reconstruct a *unique* molecular graph from them.
- ✓ **B.** Fingerprints are fixed-length *binary* (or count/numerical) vectors encoding the presence of substructures.
- ✓ **C.** Their classic uses are Tanimoto *similarity search* and as ML features.
- × **D.** They encode 2D substructure/topology, *not* 3D geometry.

Question 4: Bioassays

Select the correct statements about bioassays.

- ☐ They quantify the biological response or function that a substance triggers.
- ☐ They usually probe the substance across a range of concentrations.
- ☐ They are wet-lab procedures for characterising a compound in a biological setting.
- ☐ They are usually carried out *in vitro*, i.e. inside a living organism.

Answers

- ✓ **A.** The whole point of an assay is to quantify a biological response/function.
- ✓ **B.** They are run over a concentration range (dose–response), giving e.g. IC_{50}.
- ✓ **C.** They are physical wet-lab experiments that generate the bioactivity labels we later model.
- × **D.** The trap is the definition: *in vitro* means "in glass" (test tube / cell culture), *not* in a living organism. "In a living organism" is *in vivo*.

Question 5: ADME properties

Which statements about ADME (Absorption, Distribution, Metabolism, and Excretion) properties are correct?

- ☐ Lipinski's rule of five can serve as a proxy for ADME behaviour.
- ☐ Favourable ADME properties guarantee that a compound becomes an effective drug.
- ☐ Certain ADME properties, such as clearance, are hard to model.
- ☐ ADME properties are decisive for a drug's pharmacokinetics.

Answers

- ✓ **A.** Lipinski's Rule of Five is a cheap drug-likeness heuristic used as an ADME/oral-bioavailability proxy.
- × **B.** Necessary but *not sufficient*: good ADME means the drug reaches its target intact, but efficacy still needs actual target binding and the right pharmacodynamics.
- ✓ **C.** Properties like clearance depend on complex metabolism and are notoriously hard to predict.
- ✓ **D.** ADME *is* pharmacokinetics — what the body does to the drug over time.

Question 6: Characteristics of QSAR / bioactivity data

Which of the following correctly characterise typical QSAR / bioactivity datasets?

- ☐ QSAR datasets typically contain on the order of thousands to millions of data points.
- ☐ QSAR datasets usually show a balanced mix of active and inactive compounds.
- ☐ QSAR datasets frequently include groups of structurally related compounds.
- ☐ QSAR labels are often noisy because of biological variability and experimental inconsistency.

Answers

- × **A.** Bioactivity data is usually *small/limited* (hundreds to a few thousand per target), not millions.
- × **B.** It is typically strongly *imbalanced* — far more inactives than actives.
- ✓ **C.** Medicinal-chemistry projects explore *analog series* (one scaffold, small variations), so related compounds cluster heavily.
- ✓ **D.** Assays vary between labs/days and have measurement error, so labels are noisy.

Question 7: Applications of QSAR modeling

For which of the following applications is QSAR modeling relevant?

- ☐ Building 3D models of protein structures.
- ☐ Finding promising drug candidates.
- ☐ Estimating the toxicity of novel chemical compounds.
- ☐ Mapping the genome sequences of organisms.

Answers

- × **A.** 3D protein structure is *structure prediction* (e.g. AlphaFold), a different problem — QSAR maps molecule $\rightarrow$ property.
- ✓ **B.** QSAR ranks/scores compounds by predicted activity — i.e. finds candidates.
- ✓ **C.** Predicting properties like toxicity (QSPR) is a core QSAR application.
- × **D.** Genome sequencing/mapping is genomics, unrelated to QSAR.

Virtual Screening & QSAR Modeling

Question 8: Ligand-based virtual screening

Which statements about ligand-based virtual screening are correct?

- ☐ Molecular fingerprints are a common ingredient of ligand-based virtual screening.
- ☐ Ligand-based virtual screening cannot predict the bioactivity of untested compounds.
- ☐ Ligand-based virtual screening needs the 3D structure of the target protein.
- ☐ Ligand-based virtual screening selects compounds using bioactivity predictions alone.

Answers

- ✓ **A.** Fingerprint similarity to known actives is the workhorse of ligand-based VS.
- × **B.** It is precisely *used* to predict activity of unknown/untested compounds by analogy to known actives.
- × **C.** Needing the protein's 3D structure is *structure-based* VS. "Ligand-based" means you only use known ligands, no target structure.
- × **D.** Selection also weighs ADME, drug-likeness and structural-alert filters — not bioactivity alone.

Question 9: Data splits in QSAR model evaluation

When evaluating QSAR models, which of the following statements about data-splitting strategies hold?

- ☐ Temporal splits gauge performance on future, unseen data by mimicking how data accrue over time.
- ☐ A random split is generally the best evaluation scheme because it removes all bias.
- ☐ Cross-validation is seldom applied to QSAR because of its computational cost.
- ☐ Cluster splits keep similar compounds together so that structurally similar ones do not span the training and test sets.

Answers

- ✓ **A.** Splitting by time mimics how data really arrives and tests genuine prospective generalisation.
- × **B.** A *random* split is the classic trap: structurally similar compounds leak into both sets (compound-series bias), so it *over-estimates* performance rather than eliminating bias.
- × **C.** Cross-validation *is* routinely used in QSAR; cost is not a reason to avoid it here.
- ✓ **D.** Scaffold/cluster splits keep an entire analog series on one side, so the test set is not just memorised neighbours.

Question 10: Early-recognition metrics in virtual screening

Because only the top of a virtual-screening ranking is followed up experimentally, recognising actives early matters most. Which of the following metrics are specifically designed to reward strong early recognition?

- ☐ Precision-Recall Curve.
- ☐ BEDROC (Boltzmann-Enhanced Discrimination of ROC).
- ☐ Average Precision (AP).
- ☐ Enrichment Factor (EF).

Answers

- × **A.** A Precision–Recall curve summarises the *whole* ranking, not specifically the top.
- ✓ **B.** BEDROC applies an exponential weight that emphasises actives found *early* in the ranking.
- × **C.** Average Precision is a general ranking metric and is not early-recognition-weighted.
- ✓ **D.** The Enrichment Factor measures how many actives appear in the top $x\%$ — a pure early-recognition metric.

Question 11: Target identification / deconvolution

Which statements about target identification / deconvolution are correct?

- ☐ For a given molecule, it determines which of several simultaneously present proteins it is most likely to bind.
- ☐ It can be approached with multi-class classification.
- ☐ It can be approached with multi-task classification.
- ☐ For a given molecule, it determines every protein it would bind when several are present at once.

Answers

- ✓ **A.** Deconvolution asks: of the candidate proteins, which is the *most likely* target of this molecule.
- ✓ **B.** Picking one target out of several is naturally a *multi-class* (one-of-K) problem.
- × **C.** *Multi-task* classification predicts activity against *many* targets independently — that is target *prediction*, not deconvolution.
- × **D.** Identifying *all* binding proteins is again target prediction (many yes/no labels), not the “single most likely” framing of deconvolution.

Question 12: Drug synergy and antagonism

Which of the following statements about drug synergy and antagonism are correct?

- ☐ Synergy means the combined effect of two or more drugs exceeds the sum of their individual effects.
- ☐ The Loewe additivity model is a standard tool for assessing synergy.
- ☐ Antagonism means the combined effect of two or more drugs exceeds the sum of their individual effects.

☐ Synergy/antagonism analysis applies only to chemically similar drugs.

Answers

✓ **A.** Synergy = the combination does *more* than the sum of the parts (super-additive).

✓ **B.** Loewe additivity is the standard reference model for "no interaction", against which synergy is measured.

× **C.** This is the inverted definition: *antagonism* is when the combined effect is *less* than the sum. The sentence describes synergy but labels it antagonism.

× **D.** Combinations are studied for *any* drug pair (often deliberately different mechanisms) — chemical similarity is irrelevant.

Question 13: Random Forests vs. Graph Neural Networks

Comparing Random Forests (RF) and Graph Neural Networks (GNNs) for molecular modeling, which statements are correct?

☐ GNNs are inherently superior for QSAR because they capture the graph structure of molecules, giving more accurate predictions in every case.

☐ RF models overfit less than GNNs, which makes them more reliable on datasets with many features but few samples.

☐ RF models are usually more interpretable than GNNs, so the contribution of individual molecular features is easier to read off.

☐ GNNs are less sensitive to input-data quality and variability than RFs, making them more consistent across different datasets.

Answers

× **A.** "Better in *all* cases" is the giveaway — on limited/noisy data RF often matches or beats GNNs (no free lunch).

✓ **B.** On small, high-dimensional QSAR datasets RF (bagging) is robust and overfits less than a data-hungry GNN.

✓ **C.** RF gives feature-importance scores over interpretable descriptors; GNN representations are opaque.

× **D.** The opposite is true: GNNs are typically *more* sensitive to data quantity/quality, not less.

Graph Neural Networks & Advanced Methods

Question 14: Molecular graphs and graph neural networks

Which statements about molecular graphs and graph neural networks are correct?

☐ A molecular graph can be stored as two matrices: one for the node features and one for the edges between nodes.

☐ Node features may include the atomic number, hybridization state, and formal charge.

- ☐ A GNN uses only the node features to form its final prediction.
- ☐ Bond-type information (e.g. single vs. double bond) cannot be represented in a GNN.

Answers

- ✓ **A.** Standard encoding: a node-feature matrix plus an adjacency/edge matrix.
- ✓ **B.** Typical atom features are exactly atomic number, hybridisation, formal charge (one-hot/numeric).
- × **C.** Message passing also uses *edge* information and the graph topology, not node features alone.
- × **D.** Bond type *can* be encoded as edge features — single/double/aromatic bonds are routinely included.

Question 15: Limitations of GNNs in molecular modeling

Which of the following are genuine limitations of GNNs in the context of molecular modeling?

- ☐ GNNs cannot predict any molecular property with accuracy.
- ☐ GNNs may struggle to capture long-range interactions between atoms.
- ☐ GNNs can have trouble with very large, complex molecular graphs.
- ☐ GNNs need considerable computational resources to train.

Answers

- × **A.** Far too strong — GNNs predict many molecular properties very well; the statement is an obvious overstatement.
- ✓ **B.** A fixed number of message-passing hops limits how far information travels (over-smoothing / under-reaching).
- ✓ **C.** Large, complex graphs strain both expressivity and memory.
- ✓ **D.** Training GNNs is compute- and memory-intensive relative to e.g. RF.

Question 16: Multi-modal contrastive learning

Which statements about multi-modal contrastive learning are correct?

- ☐ Multi-modal contrastive learning embeds several modalities (e.g. molecules and proteins) into one shared latent space.
- ☐ Multi-modal contrastive learning is a self-supervised method.
- ☐ Multi-modal contrastive learning concentrates the (latent) representations of several modalities (e.g. molecules and proteins) to make predictions.
- ☐ Multi-modal contrastive learning is an unsupervised method.

Answers

- ✓ **A.** The defining idea: map both modalities into one *shared* embedding space (à la CLIP).

✓ **B.** It is *self-supervised* — supervision comes from matched/mismatched pairs, not human labels.

× **C.** It *aligns* matched pairs (pull together / push apart); "concentrates representations to make predictions" mischaracterises the objective.

× **D.** The subtle distinction: it is self-supervised, *not* unsupervised — the pairing provides the training signal.

Question 17: Multiple instance learning (MIL)

Which of the following statements about multiple instance learning (MIL) hold?

- ☐ MIL trains on sets (bags) of instances in which each instance carries its own label.
- ☐ MIL needs the bag size to be fixed.
- ☐ MIL trains on sets (bags) of instances that share a single label.
- ☐ A MIL model should be invariant to the ordering of the instances.

Answers

× **A.** Exactly what MIL is *not*: instances are *not* individually labelled (that would be ordinary supervised learning).

× **B.** Bags can have varying sizes; the aggregation (e.g. attention pooling) handles any number of instances.

✓ **C.** The hallmark of MIL: a label is attached to the *bag*, not to each instance.

✓ **D.** A bag is a set, so the model output must be *permutation-invariant*.

Generative Models & Chemical Synthesis

Question 18: Generative models for small molecules

Which statements about generative models for small molecules are correct?

- ☐ Because SMILES syntax is intricate with long-range dependencies, SMILES generators yield fewer than 10% valid strings.
- ☐ Generative models are used to explore the already-synthesised, known chemical space.
- ☐ Generative adversarial networks cannot generate small molecules.
- ☐ Autoregressive models can generate SMILES representations of small molecules.

Answers

× **A.** Empirically false: an LSTM reaches ~98% valid SMILES, and SELFIES gives 100% by construction — nowhere near "<10%".

× **B.** Generative models explore *novel* chemical space, not just the already-synthesised *known* set.

× **C.** GANs *can* generate molecules (as can VAEs, normalizing flows, diffusion).

✓ **D.** Autoregressive (next-token) models generate SMILES one character at a time and work well.

Question 19: Goal-directed molecule generation

Which of the following correctly describe goal-directed molecule generation?

- ☐ Goal-directed generation is concerned only with reproducing existing structures that already have the desired properties.
- ☐ Goal-directed generation aims to produce molecules whose properties resemble those of the training set.
- ☐ Reinforcement learning can be used to drive goal-directed molecule generation.
- ☐ Goal-directed generation aims to produce molecules with particular, pre-specified properties.

Answers

× **A.** "Replicating existing structures" describes *distribution learning*, not goal-directed design.

× **B.** "Similar to the training set" is again *distribution learning* — goal-directed deliberately pushes *beyond* the training distribution.

✓ **C.** A scoring function + RL loop (generate, score, reward, update) is a standard goal-directed recipe (e.g. REINVENT).

✓ **D.** The aim is molecules with *specific desired* properties, optimised toward an objective.

Question 20: Fréchet ChemNet Distance (FCD)

Which statements about the Fréchet ChemNet Distance (FCD) and how it is computed are correct?

- ☐ FCD measures the distance between the individual atoms of molecules.
- ☐ FCD compares the means and covariances of the molecules' latent representations.
- ☐ FCD is computed straight from the raw molecular structures with no transformation.
- ☐ FCD is a metric comparing the distribution of generated molecules with that of real ones.

Answers

× **A.** It works on whole-molecule embeddings, not distances between individual atoms.

✓ **B.** FCD fits Gaussians to ChemNet's last-hidden-layer embeddings and compares their *means and covariances.*

× **C.** It is computed in a *learned latent space* (ChemNet), not on raw structures — there is a transformation step.

✓ **D.** It is a *distributional* metric: how close is the generated set to the real set overall.

Question 21: AI for chemical reactions

Which of the following statements about machine learning for chemical reactions are correct?

- ☐ Some approaches use so-called reaction templates, i.e. transformation rules for molecules.
- ☐ Deep-learning reaction-outcome predictors can be trained on databases of known reactions.
- ☐ Transformer architectures have been applied to predicting reaction outcomes.
- ☐ Predicting reactions with machine learning requires quantum-mechanical data because of bond changes and electron movement.

Answers

- ✓ **A.** Template-based (neural-symbolic) methods select a reaction template = a transformation rule.
- ✓ **B.** Models are trained on reaction databases like USPTO (mined from patents).
- ✓ **C.** The "Molecular Transformer" treats reaction prediction as seq-to-seq translation.
- × **D.** No QM data needed — predictions come from *reaction databases*, not quantum-mechanical calculations.

Question 22: Synthesis planning

Which statements about synthesis planning are correct?

- ☐ Synthesis planning designs a sequence of chemical reactions that yields a target compound.
- ☐ Retrosynthetic analysis is central to it, decomposing complex molecules into simpler starting materials.
- ☐ Modern deep-learning methods have made synthesis planning trivial.
- ☐ Synthesis planning amounts to nothing more than choosing the reactants.

Answers

- ✓ **A.** It designs a whole *sequence* of reactions from purchasable materials to the target.
- ✓ **B.** Retrosynthesis recursively breaks the target into simpler precursors — the core technique.
- × **C.** Far from trivial: ∼50% solve rates, and whether routes actually run under real conditions remains uncertain.
- × **D.** It is much more than "selecting reactants" — it is multi-step route design with scoring and search.

Question 23: Retrosynthesis as a reinforcement-learning game

Multi-step retrosynthesis can be cast as a game solvable by reinforcement learning. Which of the following statements about this "retrosynthesis game" are correct?

- ☐ In the "game", an action consists of choosing reactants for a reaction.
- ☐ In the "game", the state is the set of available building blocks.

- ☐ In the "game", the environment carries out the reaction and returns the new molecules.
- ☐ In the "game", the reward is granted once the product has been synthesised from the building blocks.

Answers

- ✓ **A.** An *action* is choosing a disconnection / set of reactants (a reverse reaction).
- × **B.** The classic trap: the *state* is the current set of molecules *still to be made*, *not* the set of available building blocks (those are the goal you are trying to reach).
- ✓ **C.** The *environment* applies the chosen reaction and returns the new set of molecules.
- ✓ **D.** The *reward* comes when every remaining molecule is an available building block (the game is won).

Medical Data & Imaging

Question 24: Tabular medical data: types and missing values

Suppose we are given a table of patient medical records with three columns: "smoking status" (one of three string values: non-smoker, former smoker, current smoker), "cholesterol level" (floating-point measurements in mg/dL), and "diabetes risk score" (integer values in $[0, 10]$ indicating the likelihood of developing diabetes). Which of the following statements are correct?

- ☐ Missing "cholesterol level" entries can be filled with the column's median to cope with skew.
- ☐ The "smoking status" column is continuous, since it reflects patients' various smoking habits.
- ☐ "cholesterol level" is ordinal, because the measurements can be placed in increasing order.
- ☐ Missing "smoking status" entries can be filled with the column's mean to cope with skew.

Answers

- ✓ **A.** Cholesterol is numeric; the *median* is the robust choice for imputing a skewed continuous column.
- × **B.** Smoking status is *categorical* (three discrete labels), not continuous.
- × **C.** Being orderable is not enough: cholesterol is a true *continuous* (interval/ratio) measurement with meaningful differences, not merely ordinal.
- × **D.** You cannot take a *mean* of a categorical column — a missing smoking status would be imputed with the *mode*, not the mean.

Question 25: Batch effects in medical data

Which statements about batch effects in medical data are correct?

- ☐ Overlooking batch effects can yield models that separate batches instead of the true biological conditions.

- ☐ Batch effects vanish simply by collecting more samples.
- ☐ Batch effects introduce artificial differences that stem from varying experimental conditions.
- ☐ Batch effects are generally helpful, supplying needed variability to the data.

Answers

- ✓ **A.** If batches correlate with the label, the model takes the shortcut and separates *batches* instead of biology.
- × **B.** More samples from the *same* biased batches does not remove the systematic offset — you need design/correction (randomisation, ComBat, etc.).
- ✓ **C.** They are exactly artificial, non-biological differences from instruments/reagents/handling.
- × **D.** They are a confounder, not a benefit — this is the opposite of correct.

Question 26: Domain shifts in medical and healthcare data

Medical and healthcare data are frequently subject to domain shifts. Using the lecture notation (y = disease status, e.g. “healthy”/“diseased”; x = patient features, e.g. age, blood parameters, heart rate), which of the following statements about domain shifts are correct?

- ☐ A model with high predictive quality on a randomly drawn validation set will, by that fact alone, adapt to domain shifts and keep its quality over time.
- ☐ The patient-feature distribution $p(x)$ is never touched by domain shifts.
- ☐ During waves of an infectious disease (such as COVID-19), the prevalence—and thus the probability $p(y)$ of observing the phenotype—changes.
- ☐ If the way the status y is diagnosed changes, e.g. an antigen test is replaced by a PCR test, this is a concept shift $p(y \mid x)$.

Answers

- × **A.** A good *random* validation score says nothing about future shifts; models do *not* auto-adapt over time.
- × **B.** “Never” is wrong — $p(x)$ *does* shift (covariate shift), e.g. a different patient population.
- ✓ **C.** A disease wave changes the prevalence $p(y)$ — a prior/label shift.
- ✓ **D.** Redefining how the label is measured changes the $x \to y$ relationship: a concept shift $p(y \mid x)$.

Question 27: Medical imaging modalities

Which statements about medical imaging modalities are correct?

- ☐ Computed Tomography (CT) builds its body images from ultrasound.
- ☐ X-rays distinguish tissues of differing density, such as bone versus soft tissue.
- ☐ Magnetic Resonance Imaging (MRI) yields detailed images of internal structures using strong magnetic fields and radio waves.
- ☐ Optical coherence tomography forms images of organs and tissues using multiple X-rays.

Answers

- × **A.** CT is built from many *X-ray* projections, not ultrasound.
- ✓ **B.** X-ray contrast comes from differential absorption by tissue density (bone vs. soft tissue).
- ✓ **C.** MRI uses strong magnetic fields + radio-frequency pulses (no ionising radiation).
- × **D.** OCT uses *light* (near-infrared interferometry), not X-rays.

Question 28: Most common task in medical imaging

Compared with typical computer-vision tasks, which task is especially prominent in medical imaging?

- ☐ Image generation.
- ☐ Classification.
- ☐ Segmentation.
- ☐ Object detection.

Answers

- × **A.** Image generation is a niche/auxiliary use, not the dominant clinical task.
- ✓ **B.** Classification (e.g. disease present/absent, grading) is also very common.
- ✓ **C.** *Segmentation* (delineating organs/lesions pixel-by-pixel) is the signature, disproportionately common medical-imaging task — hence the U-Net.
- × **D.** Object detection is comparatively rare in medical imaging versus segmentation and classification.

Question 29: The U-Net architecture

Which of the following correctly describe the U-Net architecture?

- ☐ U-Net was the first vision model whose encoder relied on self-attention rather than convolutions.
- ☐ The U-Net architecture has a symmetric encoder and decoder.
- ☐ U-Net was created specifically for the object-detection task.
- ☐ To help decoding, encoder feature maps are concatenated with those of the matching up-sampling level of the decoder.

Answers

- × **A.** U-Net is fully *convolutional*; it predates and does not use self-attention encoders.
- ✓ **B.** The “U” is a symmetric contracting encoder + expanding decoder.
- × **C.** It was designed for *segmentation* (biomedical), not object detection.
- ✓ **D.** Skip connections concatenate encoder feature maps into the matching decoder level — the key idea that preserves spatial detail.

Question 30: The "Clever Hans" effect

In the context of AI for medical imaging, what does the "Clever Hans" effect refer to?

- ☐ A bias that arises when up-scaling the resolution of low-quality medical images.
- ☐ A case where a model looks like it performs well but is actually leaning on irrelevant features or artifacts in the data.
- ☐ A case where an AI predicts outcomes incorrectly because its training data were biased.
- ☐ The use of data augmentation to make medical-imaging models more robust.

Answers

- × **A.** Unrelated to super-resolution bias.
- ✓ **B.** Exactly shortcut learning: the model *looks* accurate but is exploiting an irrelevant artifact (e.g. a hospital tag/ruler), not the pathology.
- × **C.** Tempting, but too generic: "biased training data" describes *any* bias. Clever Hans specifically means relying on *spurious shortcut features* while appearing to perform well.
- × **D.** Data augmentation is a mitigation technique, not the Clever Hans effect.

Question 31: Explainable AI and interpretability in medical imaging

Which statements about explainable AI and interpretability in medical imaging are correct?

- ☐ Gradient integration propagates a relevance value backwards through the layers to the input, estimating each input feature's relevance.
- ☐ Explainable-AI methods such as Grad-CAM help visualise which parts of a medical image matter most for a prediction.
- ☐ Occlusion analysis deletes patches from the training images to make the model more interpretable.
- ☐ Counterfactual explanations generate alternative images showing what would have had to change for a different diagnosis.

Answers

- × **A.** The layer-by-layer back-propagation of a *relevance* value is Layer-wise Relevance Propagation (LRP); "gradient integration" is the wrong name for it.
- ✓ **B.** Grad-CAM produces a heatmap of the image regions most responsible for the prediction.
- × **C.** Occlusion perturbs the *input* image at inference (mask a patch, see the score drop) — it does *not* remove training data, and it explains rather than "improves interpretability".
- ✓ **D.** Counterfactuals show the minimal image change that would flip the diagnosis — a what-if explanation.

Question 32: AlphaFold 2

Which of the following statements about AlphaFold 2 are correct?

- ☐ AlphaFold 2 relies on quantum-mechanical calculations to sharpen its predictions.

☐ AlphaFold 2 draws on multiple sequence alignments and homologous protein structures to improve its accuracy.

☐ AlphaFold 2 is a deep-learning model that predicts a protein's three-dimensional structure from its amino-acid sequence.

☐ AlphaFold 2 is built on a convolutional neural network architecture.

Answers

× **A.** No quantum-mechanical calculations are used — it is a learned model.

✓ **B.** It relies heavily on multiple sequence alignments (co-evolution signal) and template structures.

✓ **C.** Its job is exactly sequence → 3D structure prediction.

× **D.** The architecture is *attention*-based (the Evoformer + structure module), not a CNN.

Biological Sequences

Question 33: Composition of biological sequences

Which statements about the composition of biological sequences are correct?

☐ Protein sequences are chains of amino acids joined by peptide bonds.

☐ DNA sequences are built from nucleic acids.

☐ Biological sequences obey no particular patterns or rules in their make-up.

☐ RNA sequences are built from lipids.

Answers

✓ **A.** Proteins are amino-acid chains joined by peptide bonds.

✓ **B.** DNA is a chain of nucleotides (nucleic acid).

× **C.** Sequences are highly structured (motifs, codons, grammar) — "no rules" is false and is what makes them learnable.

× **D.** RNA is a *nucleic acid*, not a lipid.

Question 34: Deep learning for biological sequences

Which of the following statements about deep learning methods for biological sequences are correct?

☐ Feature extraction from biological sequences can reveal motifs, i.e. recurring patterns or elements within them.

☐ One-hot encoding is a common way to prepare biological sequences as input for deep models.

☐ Deep-learning techniques apply to tasks such as classification, segmentation, and generative modelling of biological sequences.

- ☐ 1D convolutions, Recurrent Neural Networks (RNNs), Long Short-Term Memory networks (LSTMs), and Transformers are all used to analyse biological sequences.

Answers

- ✓ **A.** Learned filters detect recurring *motifs* (e.g. binding sites).
- ✓ **B.** One-hot encoding of the alphabet (A/C/G/T or 20 amino acids) is the standard input format.
- ✓ **C.** Classification, segmentation (per-position labels) and generation are all applied to sequences.
- ✓ **D.** 1D-CNNs, RNNs, LSTMs and Transformers are all standard sequence architectures here — every option is correct.

2026 exam — single-best-answer

The remaining sections are from the **2026** exam (25 June 2026). Each question has **exactly one** correct option among the three listed; the answer box marks it ✓ and explains why the other two are wrong.

Molecules, Representations & Cheminformatics (2026)

Question 35: Molecular graphs vs. fingerprints

Which of the following statements accurately describes the relationship between molecular graphs and molecular fingerprints?

- ☐ A fingerprint can be computed from a molecular graph, yet the exact graph usually cannot be recovered from the fingerprint.
- ☐ Fingerprints and molecular graphs must hold the same structural information.
- ☐ A molecular graph can always be reconstructed exactly from its fingerprint.

Answers

- ✓ **A.** A fingerprint is *computed from* the graph (substructure enumeration/hashing), but the mapping is lossy and many-to-one, so the exact graph cannot be recovered.
- × **B.** A fingerprint is a compressed bit/count summary that discards information, so it does *not* carry the same structural information as the graph.
- × **C.** Exact reconstruction is impossible in general — different molecules can even hash to the same bits.

Question 36: Converting between SMILES and fingerprints

Using standard cheminformatics tools, in which direction is conversion between a SMILES string and a molecular fingerprint actually possible?

- ☐ Cheminformatics tools can convert SMILES and fingerprints into one another in both directions.

- ☐ Cheminformatics tools can turn a SMILES into a fingerprint, but not the other way round.
- ☐ Cheminformatics tools can turn a fingerprint into a SMILES, but not the other way round.

Answers

× **A.** The round-trip does not exist: you cannot go fingerprint → SMILES.

✓ **B.** SMILES → molecular graph → fingerprint is a standard cheminformatics pipeline.

× **C.** Backwards — fingerprints are lossy, so fingerprint → SMILES is not possible.

Question 37: Tanimoto coefficient

On what quantity is the Tanimoto coefficient based when it scores how similar two molecular fingerprints are?

- ☐ It takes the count of shared features over the total number of features present in either molecule.
- ☐ It superimposes three-dimensional pharmacophore points and averages their spatial distances.
- ☐ It compares the core scaffolds, scoring one whenever the scaffolds coincide.

Answers

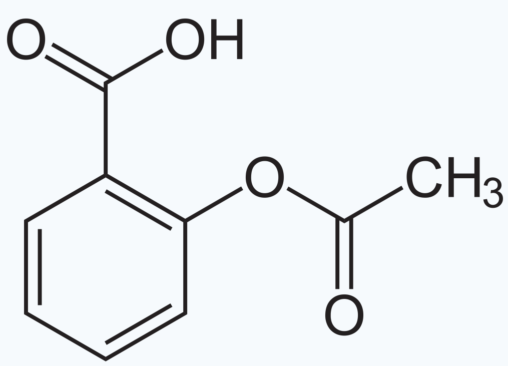
✓ **A.** Tanimoto $= |A \cap B|/|A \cup B|$: shared on-bits divided by the union of on-bits.

× **B.** Averaging 3D pharmacophore-point distances is a different (3D alignment) similarity, not Tanimoto.

× **C.** Tanimoto does not test scaffold identity; it counts shared fingerprint features.

Question 38: Aspirin SMILES

Which of the following SMILES strings represents the molecule shown below?



- ☐ `O=C(C)Oc1ccccc1C(=O)O`
- ☐ `cc(=O)Oc1ccccc1c(=O)O`
- ☐ `O=C[C]Oc1ccccc1C[=O]O`

Answers

✓ **A.** `O=C(C)Oc1ccccc1C(=O)O`: an acetyl ester (`O=C(C)O`) on an aromatic ring (`c1ccccc1`) bearing a carboxylic acid (`C(=O)O`) — acetylsalicylic acid.

× **B.** Lowercase `c` marks *aromatic* atoms, so writing the carbonyl carbons as aromatic (`cc(=O)`) is chemically and syntactically wrong.

× **C.** `[C]` and `[=O]` are malformed SMILES atom tokens — the string does not parse.

Question 39: Bioactivity datasets

Which of the following best captures what bioactivity datasets usually look like in practice?

- ☐ They are usually large, balanced, and almost free of measurement noise.
- ☐ They are frequently small, imbalanced, and noisy, with far more inactive than active compounds.
- ☐ They are cheap to produce and therefore cover most compound–assay combinations.

Answers

× **A.** The opposite of reality — bioactivity data is rarely large, balanced, or clean.

✓ **B.** It is typically limited, imbalanced (far more inactives than actives), and noisy.

× **C.** Assays are expensive, so coverage of compound×assay combinations is very sparse.

Question 40: The Design–Make–Test–Analyse cycle

In drug discovery, which option correctly captures how the Design–Make–Test–Analyse (DMTA) cycle proceeds?

- ☐ Compounds are tested before being designed, and only the active ones are afterwards synthesized.
- ☐ Compounds are designed and made just once, after which testing removes any need for further design.
- ☐ Compounds are designed, made, and tested experimentally, and the analysis steers the next design round.

Answers

× **A.** Designing comes before testing — you cannot test a compound you have not yet designed and made.

× **B.** It is an iterative *cycle*; one round of testing does not remove the need for further design.

✓ **C.** Design → Make → Test → Analyse, and the analysis feeds the next design iteration.

Virtual Screening, QSAR & Targets (2026)

Question 41: Structure-based vs. ligand-based drug design

What is the essential difference between structure-based drug design (SBDD) and ligand-based drug design (LBDD)?

- ☐ SBDD needs known active ligands, whereas LBDD needs the target's 3D structure.
- ☐ Both SBDD and LBDD need a high-resolution 3D structure of the target.
- ☐ SBDD relies on the target's 3D structure, whereas LBDD can work from known ligands without that structure.

Answers

- × **A.** Swapped: SBDD needs the target structure, LBDD needs known ligands.
- × **B.** LBDD specifically does *not* require the target's 3D structure.
- ✓ **C.** SBDD uses the target's 3D structure; LBDD relies on known ligands without it.

Question 42: Similarity search vs. model-based search

In what way do similarity search and model-based search differ from each other?

- ☐ Similarity search is ligand-based, whereas model-based search must have the 3D structure of the target's binding site.
- ☐ Similarity search retrieves molecules resembling a query, whereas model-based search predicts activity with a model trained on labelled data.
- ☐ Similarity search learns a decision function from labelled actives and inactives, whereas model-based search ranks compounds by closeness to a query.

Answers

- × **A.** Model-based search does *not* necessarily need the target's 3D structure — it can be a ligand-based supervised model.
- ✓ **B.** Similarity search retrieves neighbours of a query molecule; model-based search predicts activity with a model trained on labels.
- × **C.** The two definitions are reversed.

Question 43: Early-recognition metrics (odd one out)

Of the metrics below, which one is *not* built to reward finding actives early, i.e. near the top of a ranked virtual-screening list?

- ☐ BEDROC.
- ☐ AUC-ROC.
- ☐ Enrichment factor.

Answers

- × **A.** BEDROC is built specifically to reward early recognition.
- ✓ **B.** AUC-ROC weights all ranking positions equally, so it is *not* an early-recognition

metric — the odd one out.

× **C.** The enrichment factor measures actives in the top $x\%$ — an early-recognition metric.

Question 44: Evaluating a virtual-screening model

A virtual-screening model orders a million compounds, yet your budget only permits assaying the top 1%. Which model-selection strategy best fits this situation?

☐ Pick the model with the best accuracy at a 0.5 probability threshold, since ranking above that threshold no longer matters once compounds are classified.

☐ Pick the model with the highest overall AUC-ROC, since weighting all ranking positions equally guarantees the best enrichment in the top 1%.

☐ Favour an early-recognition metric such as the enrichment factor near the 1% cutoff or BEDROC, since the top of the ranking decides experimental usefulness.

Answers

× **A.** A 0.5-threshold accuracy ignores how well the very top of the ranking is ordered — which is all that matters here.

× **B.** Global AUC-ROC does *not* guarantee good enrichment inside the top 1%.

✓ **C.** Use an early-recognition metric — enrichment factor near the 1% cutoff, or BEDROC.

Question 45: Defensible generalization estimates in QSAR

In a QSAR pipeline, which procedure yields the most trustworthy estimate of how well the model generalises?

☐ Fix the split first, carry out feature and hyperparameter selection on the training data, and reserve the held-out test set for the final evaluation only.

☐ Split first, then compare models and hyperparameters on the test set and report the best test-set score.

☐ Select features on the whole dataset, then split the reduced data and tune hyperparameters on the training part only.

Answers

✓ **A.** Split first, do all feature/hyperparameter selection on the training data, and touch the test set only once for the final estimate.

× **B.** Tuning on the test set leaks it into model selection — the reported number is optimistic.

× **C.** Selecting features on the full dataset before splitting also leaks test information.

Question 46: Compound-series bias

What does compound-series bias refer to, and why does it undermine QSAR model evaluation?

☐ Because many compounds are closely related, a random split can put near-analogues in both training and test, underestimating the generalization error.

- ☐ Test-compound information leaks into feature or hyperparameter selection, biasing the reported score even when compounds are structurally unrelated.
- ☐ The active class is far smaller than the inactive one, so the majority class dominates performance even when the chemical series are diverse.

Answers

- ✓ **A.** Close analogues land in both train and test under a random split, so the test set is partly memorised and generalization error is underestimated.
- × **B.** That describes generic information leakage in feature/hyperparameter selection, not compound-series bias.
- × **C.** That describes class imbalance, a different problem.

Question 47: Drug synergy and model symmetry

Which option combines a correct definition of drug synergy with a sensible property that a model predicting it should have?

- ☐ The combined effect is larger than the additive one, and the prediction should stay the same when the two compounds are swapped.
- ☐ The combined effect equals the additive expectation, and the prediction may differ when the pair is supplied in reverse order.
- ☐ The combined effect is below the additive expectation, and the prediction should be unaffected by the order of the compounds.

Answers

- ✓ **A.** Synergy is super-additive (combined > additive), and a sensible model is symmetric — swapping the two compounds must not change the prediction.
- × **B.** "Equals the additive expectation" is *no* interaction, and order-dependence is an undesirable property.
- × **C.** "Smaller than additive" is *antagonism*, not synergy (the symmetry property is right, but the definition is wrong).

Question 48: Proteochemometric modelling

In what respect does proteochemometric modelling go beyond classical single-target QSAR?

- ☐ It uses representations of both the molecule and the protein target together.
- ☐ It uses molecular representations with a softmax over the candidate protein targets.
- ☐ It uses molecular representations with one separate prediction head per protein target.

Answers

- ✓ **A.** Proteochemometrics jointly encodes the *molecule and the protein target*, so one model spans many targets.
- × **B.** A softmax over targets from the molecule alone has no target representation — not the proteochemometric idea.

× **C.** A separate head per target (multi-task) still uses only molecular inputs — not a joint molecule+protein representation.

Question 49: Target deconvolution / target fishing

How is the task known as target deconvolution (or target fishing) defined?

☐ Given one protein and a compound library, rank every molecule by predicted affinity to that protein.

☐ Given one molecule and several candidate proteins, predict which protein it is most likely to bind.

☐ Given one molecule, predict binding to each candidate protein independently, so any number of targets may be assigned at once.

Answers

× **A.** Ranking a whole library against *one* protein is virtual screening — the inverse problem.

✓ **B.** Target fishing: given a molecule and several candidate proteins, identify the *most likely* target.

× **C.** Predicting binding to *every* target independently is multi-task target prediction, not deconvolution's "single most likely target" framing.

Graph Neural Networks (2026)

Question 50: Reordering atoms in a molecular graph

Suppose the atoms of a molecular graph are given new indices while the bonds between them are left unchanged. What behaviour do we expect from a graph neural network?

☐ The node representations are permuted accordingly, but the graph-level prediction may change because the adjacency matrix was reordered.

☐ Both the node representations and the graph-level prediction stay in exactly the same tensor positions.

☐ The node representations are permuted accordingly, while the graph-level prediction stays unchanged.

Answers

× **A.** A pure relabelling must *not* change the graph-level prediction — it is the same graph, just reindexed.

× **B.** Node representations should follow the permutation, not stay in fixed positions.

✓ **C.** Correct symmetry: node features are *equivariant* (they permute with the atoms) while the graph-level readout is *invariant*.

Question 51: Steps of an MPNN layer

What is the usual order of operations carried out inside a single Message Passing Neural Network (MPNN) layer?

- ☐ Generate messages, aggregate them, then update.
- ☐ Generate messages, apply graph convolution, then aggregate.
- ☐ Apply graph convolution, generate messages, then update.

Answers

- ✓ **A.** One MPNN layer: generate messages, aggregate them at each node, then update the node state.
- × **B.** "Graph convolution" is not a separate sub-step here, and aggregation is not the final step.
- × **C.** Wrong order and wrong steps — the update must come last.

Question 52: Oversmoothing in GNNs

Once a large number of message-passing layers are stacked, the node embeddings across a molecular graph end up nearly identical. Why does this cause trouble?

- ☐ Long-range information is squeezed through narrow graph bottlenecks, so distant signals cannot travel effectively.
- ☐ The model may no longer tell local atom environments apart because the node embeddings grow too similar.
- ☐ Important information never reaches distant nodes because too few message-passing steps were taken.

Answers

- × **A.** Distant signals failing through bottlenecks is *over-squashing*, a different pathology.
- ✓ **B.** Oversmoothing: after many layers the node embeddings converge, so the model can no longer tell local atom environments apart.
- × **C.** "Too few steps" (under-reaching) is the opposite of the "many layers" premise.

Question 53: Contrastive learning

By what mechanism does contrastive learning build a joint molecule–target embedding space?

- ☐ It clusters molecules and targets separately and assumes equal-index clusters are binding pairs.
- ☐ It pulls associated molecule-target pairs closer together and pushes mismatched pairs apart.
- ☐ It concatenates molecule and target representations and fits a supervised activity regressor without comparing pairs in a shared space.

Answers

× **A.** Clustering each modality separately and matching by cluster index is not contrastive learning.

✓ **B.** Contrastive learning pulls associated molecule–target pairs together and pushes mismatched pairs apart in a shared space.

× **C.** Concatenating and fitting a supervised regressor never builds a shared comparison space.

Question 54: Multiple Instance Learning

How does Multiple Instance Learning depart from ordinary supervised learning, in which every feature vector carries a single label $(\boldsymbol{x}, y)$?

- ☐ . . . we are given a bag of feature vectors that share a single label $(\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}, y)$.
- ☐ . . . we are given a bag of feature vectors paired with a bag of labels $(\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}, \{y_1, \ldots, y_M\})$.
- ☐ . . . we are given single feature vectors paired with a bag of labels $(\boldsymbol{x}, \{y_1, \ldots, y_M\})$.

Answers

✓ **A.** MIL replaces $(\boldsymbol{x}, y)$ with a *bag* of vectors sharing one label, $(\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}, y)$.

× **B.** A bag labelled with a *bag* of values is not MIL (that would be a multi-output/multi-label setup).

× **C.** A single vector with multiple labels is ordinary multi-label learning, not MIL.

Generative Models & Synthesis Planning (2026)

Question 55: Reward hacking a QSAR proxy

A generator is trained to maximise a QSAR-based activity score and learns to output molecules the score rates as highly active, yet later wet-lab assays find them inactive. What most plausibly went wrong?

- ☐ The activity score was supplied only after each molecule was finished, so the reward was delayed.
- ☐ The generator gamed weaknesses in the proxy scoring model instead of raising genuine experimental activity.
- ☐ The generator made too few structurally distinct molecules, lowering the diversity of candidates.

Answers

× **A.** Delayed (sparse) reward does not explain why *predicted* activity is high while *real* activity is low.

✓ **B.** Classic reward hacking: the generator found inputs that fool the proxy QSAR model instead of being genuinely active.

× **C.** Low diversity would not systematically inflate predicted activity above true activity.

Question 56: Combating analogue rediscovery (odd one out)

A goal-directed generator keeps producing close analogues of molecules it has already scored highly. Which of the following would *not* help to counteract this tendency?

- ☐ Blend samples from a fixed exploratory generator with those of the trainable generator while generating SMILES.
- ☐ Penalise the reward of molecules resembling earlier high-scoring ones kept in a memory buffer.
- ☐ Push exploitation harder by always taking the highest-scoring molecule as the next training example.

Answers

- × **A.** Mixing in a fixed exploratory generator adds diversity — a genuine remedy, so not the answer.
- × **B.** Penalising similarity to a memory of past hits also promotes diversity — a genuine remedy.
- ✓ **C.** This is the answer: always taking the single highest-scoring molecule *increases* exploitation and worsens rediscovery, so it is *not* a remedy.

Question 57: Autoregressive molecular reinforcement learning

When molecule construction is performed autoregressively and treated as a reinforcement-learning problem, which assignment of the RL roles is correct?

- ☐ The partial molecule is the state and the next building step is the action, but the scoring function is the policy and the generator gives the reward.
- ☐ The finished molecule is the state, the scoring function is the action, and each partial molecule gets its own final reward.
- ☐ The partial molecule is the state, the next building step is the action, the generator is the policy, and the finished molecule earns the reward.

Answers

- × **A.** The *generator* is the policy, not the scoring function, and the generator does not "supply the reward".
- × **B.** The completed molecule is not the state, and partial molecules do not each get an independent final reward.
- ✓ **C.** Partial molecule = state, next construction step = action, generator = policy, completed molecule earns the reward.

Question 58: Diffusion vs. one-shot latent decoding

What sets diffusion-based molecular generation apart from decoding a molecule from a single latent vector in one pass?

- ☐ It maps a single latent vector straight to a finished molecule in one decoder pass.
- ☐ It begins from noise and repeatedly applies a learned denoising process.

- ☐ It builds the molecule step by step, predicting the next token or graph action from the partial molecule.

Answers

- × **A.** Mapping one latent vector to a full molecule in a single pass is exactly one-shot decoding.
- ✓ **B.** Diffusion starts from noise and *iteratively* denoises toward a molecule.
- × **C.** Step-by-step next-token / graph-action generation is autoregressive, not diffusion.

Question 59: Conditional generation

Trained on molecule–logP pairs, a model takes a target logP value as input and emits molecules in a single forward pass, never querying a logP scoring function during generation. Which generative paradigm is this?

- ☐ Conditional generation.
- ☐ Goal-directed generation.
- ☐ Distribution learning followed by unconditional sampling.

Answers

- ✓ **A.** The property is given as a *condition*, the model maps it to molecules in one pass, and no scoring function is queried — conditional generation.
- × **B.** Goal-directed generation optimises against a scoring function during generation, which is explicitly excluded here.
- × **C.** Unconditional sampling would ignore the desired logP input.

Question 60: Distribution learning

In molecular generation, which option most accurately captures what distribution learning is?

- ☐ Distribution learning produces molecules conditioned on side information such as protein targets or desired property values.
- ☐ Distribution learning seeks to produce molecules resembling the training set by learning its underlying distribution.
- ☐ Distribution learning concentrates on producing molecules optimised for specific biological or physicochemical properties.

Answers

- × **A.** Conditioning on auxiliary information is *conditional* generation.
- ✓ **B.** Distribution learning models the training distribution and samples molecules resembling it.
- × **C.** Optimising for specific properties is *goal-directed* generation.

Question 61: Mode collapse

In a generative model for molecules, what does the term mode collapse refer to?

- ☐ The model produces molecules that are valid and novel but match the training-set property distribution poorly.
- ☐ The model keeps emitting the same or nearly the same molecules.
- ☐ The model emits almost only molecules that already appear in the training set.

Answers

- × **A.** Good validity/novelty but poor distribution match is a different failure mode, not mode collapse.
- ✓ **B.** Mode collapse: the generator keeps producing the same or very similar molecules (it covers only a few "modes").
- × **C.** Re-emitting training molecules is memorisation, not mode collapse.

Question 62: Reaction prediction and retrosynthesis

Which of the options correctly pairs each task with its corresponding inputs and outputs?

- ☐ Both forward prediction and single-step retrosynthesis take only the reaction conditions and predict the yield.
- ☐ Forward prediction maps reactants to products, while single-step retrosynthesis maps a target product to candidate precursors.
- ☐ Forward prediction maps products to reactants, while single-step retrosynthesis predicts the major product from given reactants.

Answers

- × **A.** Neither task takes only conditions and predicts yield.
- ✓ **B.** Forward prediction: reactants $\rightarrow$ products; single-step retrosynthesis: product $\rightarrow$ precursors.
- × **C.** The two directions are swapped.

Question 63: Reaction templates

What role is a reaction template intended to serve?

- ☐ To give a general symbolic rule stating the required substructures and how reactants turn into products.
- ☐ To capture reaction conditions such as solvent and temperature, leaving the structural change to a neural model.
- ☐ To attach a reaction-class label grouping similar reactions, without saying how atoms or bonds change.

Answers

- ✓ **A.** A reaction template is a generalised symbolic rule: required substructures plus how reactants become products.
- × **B.** Encoding solvent/temperature conditions is not what a template captures.
- × **C.** Assigning a reaction *class* without the atom/bond transformation is classification, not a template.

Question 64: Easiest reaction task

Among the tasks listed below, which one is typically the least difficult?

- ☐ Single-step retrosynthesis.
- ☐ Multi-step retrosynthesis planning.
- ☐ Forward reaction prediction.

Answers

- × **A.** Single-step retrosynthesis is one-to-many (many valid precursor sets), so it is harder.
- × **B.** Multi-step planning searches over many steps — the hardest of the three.
- ✓ **C.** Forward reaction prediction is the most constrained (reactants $\rightarrow$ major product) and generally the easiest.

Question 65: Template-free translation methods

In what terms do template-free translation methods formulate reaction prediction or retrosynthesis?

- ☐ As looking up the most compatible stored reaction pattern in an associative memory.
- ☐ As sorting the input into a learned reaction category and then applying a symbolic transformation.
- ☐ As a direct sequence-to-sequence translation between molecular reaction strings.

Answers

- × **A.** Retrieving stored patterns from memory is a template/retrieval approach.
- × **B.** Classifying into a learned reaction class then transforming is template-based.
- ✓ **C.** Template-free methods translate reaction strings directly, as sequence-to-sequence (e.g. the Molecular Transformer).

Question 66: Deep networks inside MCTS for retrosynthesis

What makes deep neural networks helpful when they are embedded within Monte Carlo Tree Search for retrosynthesis?

- ☐ They take over the search by scoring only the first disconnection and then running that route greedily.
- ☐ They can estimate the value of unvisited states and propose promising actions when exhaustive search is impractical.

- ☐ They mostly balance chemical equations at each node, while the search alone fixes value and policy without learned guidance.

Answers

- × **A.** They do not collapse the search to a single greedy first disconnection.
- ✓ **B.** Learned networks estimate the value of unvisited states and propose promising actions, guiding the search where exhaustive enumeration is infeasible.
- × **C.** They provide the value/policy guidance — the search is not left unguided.

Question 67: Pocket-conditioned 3D ligand generation

Which option most accurately describes pocket-conditioned 3D ligand generation?

- ☐ Ligands are built without protein context and only docked afterwards; the docking score does not steer the generator.
- ☐ The protein pocket supplies geometric context for building or denoising a ligand, usually followed by validity, docking, property, and synthesis checks.
- ☐ A fixed ligand graph is supplied and only different conformations of that same molecule are sampled, ignoring the protein.

Answers

- × **A.** Generating without protein context and only docking afterwards is *not* pocket-conditioned — the pocket never conditions the generator.
- ✓ **B.** The binding pocket gives geometric context to generate/denoise the ligand, typically followed by validity, docking, property, and synthesis checks.
- × **C.** Sampling conformers of a fixed ligand without protein information is not pocket-conditioned generation.